DSA06 DATA HANDLING AND VISUALIZATION | Activity: Correlation Detective | Slot C

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Time: 20 to 25 minutes Total: 20 marks Tool: R (ggplot2 / base R). Work alone; your case file is different from your neighbour's. Write your answers and paste your plots and code.

Case File CD-C04: The Dairy farm Case [20 marks]

Briefing: You are the data detective hired by the manager of a dairy farm. The file holds 16 monthly records from the monsoon year, with seven variables. Some relationships can be trusted. Others are traps: a hidden driver, a curve that fools r, and a single wrong record. Find them.

R name	What it measures	R name	What it measures
rainfall	Rainfall (mm)	crop_damage	Crop damage (%)
grass_height	Grass height (cm)	tractor_hours	Tractor hours
milk_yield	Milk yield (litres per cow)	fuel_cost	Fuel cost (thousand rupees)
fert_dev	Fertiliser minus recommended dose (kg per ha)		

Evidence: run this code in R to create your case file (do not edit it).

```
set.seed(225976)
n <- 16
rainfall <- round(rnorm(n, 105.9, 47.4), 1)
grass_height <- round(30.9 + 10.7 * (0.96 * (rainfall - 105.9) / 47.4 + 0.28 * rnorm(n)), 1)
milk_yield <- round(12.4 + 3.6 * (0.87 * (rainfall - 105.9) / 47.4 + 0.48 * rnorm(n)), 1)
fert_dev <- round(seq(-32.9, 32.9, length.out = n)[order(runif(n))], 1)
crop_damage <- round(5.3 + 19.0 * (fert_dev / 32.9)^2 + rnorm(n, 0, 0.7), 1)
tractor_hours <- round(rnorm(n, 79, 23.7), 0)
fuel_cost <- round(39.7 + 11.9 * (0.17 * (tractor_hours - 79) / 23.7 + 0.99 * rnorm(n)), 1)
tractor_hours[10] <- 158; fuel_cost[10] <- 76.5   # row 10 was typed in by a trainee
case <- data.frame(rainfall, grass_height, milk_yield, fert_dev, crop_damage, tractor_hours, fuel_cost)
```

Tasks:

1. The matrix. Use cor(case) to get the 7 x 7 correlation matrix (round to 2 decimals). State n and how many distinct pairs of variables there are. Draw a correlogram of the matrix (ggplot2 geom_tile with geom_text, or corrplot). [4]
2. The strongest clues. Name the strongest positive pair and the strongest negative pair (off-diagonal). Give r for each and call it weak (below 0.3), moderate (0.3 to 0.7) or strong (above 0.7). [3]
3. The deceptive zero. Plot crop_damage against fert_dev with a straight line, then add a quadratic fit (formula = y ~ x + I(x^2)). Report r and the R-squared of the quadratic fit. In one sentence, say why r misses this relationship. [4]
4. The outlier alibi. Plot fuel_cost against tractor_hours. Find the row that looks wrong and give its row number. Report r for all rows and r without that row, and say what the single record did to r. [4]
5. The prime suspect. grass_height and milk_yield are strongly linked. Report r(grass_height, milk_yield), then r(rainfall, grass_height) and r(rainfall, milk_yield). Compute the partial correlation of grass_height and milk_yield controlling for rainfall: correlate resid(lm(grass_height ~ rainfall)) with resid(lm(milk_yield ~ rainfall)). Write a two-sentence verdict: does grass_height cause milk_yield? Name one study design that could test causation. [5]

Detective's rule: *correlation measures a straight-line link only, it can be inflated by one point, and it never proves cause.*

Code of Conduct: I certify that this work is my own. Signature: _____

Activity-7

1. Correlation matrix

observations : n = 16

Variables : 7

Distinct pairs : 7(7-1)/2 = 21

```
c <- cor(case)
round(c,2)
library(ggplot2)
d <- as.data.frame(as.table(c))
ggplot(d, aes(Var1, Var2, fill=Freq)) +
  geom_tile() +
  geom_text(aes(label = round(Freq,2))).
```

2. strongest clues.

```
c <- cor(case)
c[lower.tri(c, diag=TRUE)] <- NA
which(c==max(c, na.rm=TRUE), arr.ind=True)
which(c==min(c, na.rm=TRUE), arr.ind=True)
```

Interpretation:- The strongest positive pair is expected to be rainfall and grass height. use the output to identify the strongest negative pair and its exact r.

3. The deceptive zero.

Plot (crop.damage ~ pest.dev, data = case)

abline (lm(crop.damage ~ pest.dev, case), col = "b

m <- lm (crop.damage ~ pest.dev + I (pest.dev ^ 2))

x <- sort (pest.dev)

lines (x, Predict (m, data.frame (pest.dev = x)), col =

cor (case $ crop.damage, case $ pest.dev)

summary (m) $ r.squared.

4. The outlier alibi

Plot (fuel.cost ~ tractor.hours, data = case)

text (case $ tractor.hours, case $ fuel.cost, labels

= 1:16, pos = 3)

cor (case $ tractor.hours, case $ fuel.cost)

cor (case $ tractor.hours [-10], case $ fuel.cost

[-10]).

5. The prime suspect.

cor (case $ grass.height, case $ milk.yield).

cor (case $ rainfall, case $ grass.height)

cor (case $ rainfall, case $ milk.yield)

cor (resid (lm (grass.height ~ rain fall, case)),

resid (lm (milk.yield ~ rainfall, case))).